

# Ruiyang Wang

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## EDUCATION

### University of California, Los Angeles (UCLA)

Los Angeles, CA

*Master of Engineering, Artificial Intelligence*

Fall 2026 – Summer 2027

- Relevant Coursework: Neural Networks and Deep Learning (ECE C247A), Current Topics in Artificial Intelligence (CS 269), Management Communication (ENGR 214)

### California State Polytechnic University, Pomona (Cal Poly Pomona)

Pomona, CA

*Bachelor of Science, Computer Science, Minor in Data Science*

Fall 2022 – Spring 2026

- President's Honor List (2023-2025), Dean's List (2023-2025)
- Engage Student Research Stipend (\$2,000), 2025-2026

## EXPERIENCE

### NASA Jet Propulsion Laboratory (JPL) Internship

Pasadena, CA, May 2026 - June 2026

- Cleared competitive technical selection to integrate AI/ML into mission scheduling pipeline for flagship deep-space mission (Europa Clipper); onboarding paused due to federal funding/clearance restructuring

### NASA L'SPACE, Proposal Writing and Evaluation Experience, PI

Online, January 2026 - April 2026

- Selected as Principal Investigator from 50+ nationwide submissions; led 8-member team spanning 8 universities designing "Tumbling Prediction for Non-Cooperative Space Debris" mission; coordinated with SME (Dr. Sumant Sharma, EraDrive CEO).
- Designed physics-informed ML architecture (Python, TensorFlow/Keras) achieving 67% accuracy improvement ( $\leq 5^\circ$  vs.  $\sim 15^\circ$  attitude error),  $5\times$  faster operation, enabling autonomous capture planning

### Cal Poly Pomona, Research Fellow

Pomona, CA, September 2024 - May 2026

*Engage Program: AI & Crowdfunding Trust · Aug 2025 – May 2026*

- Analyzed AI-generated linguistic features' influence on investor trust via logistic regression on 220 Crowdcube campaigns; identified signals associated with  $2.7\times$  higher survival probability ( $p=0.018$ )
- Found sustainability positioning timing achieves  $8.6\times$  better survival outcomes; presented 48"×36" poster at Cal Poly RSCA Conference to 100+ interdisciplinary attendees

*Anomaly Detection (MPI & Deep Learning) · Sep 2024 – May 2025*

- Built modular PyTorch deep learning pipeline for network anomaly detection on 5.9M-sample CIC-IDS2017 dataset; parallelized inference with MPI (mpi4py) across 32 processes achieving  $\sim 217\times$  speedup
- Implemented MPI collective operations (MPI\_Bcast, MPI\_Gather) reducing communication overhead  $\sim 40\%$ ; benchmarked 4–64 processors validating Amdahl's Law, ensured reproducibility via fixed seeds ( $< 1e-6$  error)

## LEADERSHIP

### LoopBreakers Technical Team, Founder & Team Lead

April 2025 - Present

- Built **Uptime!**, an AI-powered sitting reminder Chrome extension (Manifest V3, Vanilla JS) integrating Gemini API for personalized wellness nudges and AI-generated fun rewards; deployed landing page on Vercel (Next.js); published on Chrome Web Store
- Led 4-person team building full-stack wildfire alert AI agent (NASA FIRMS API + Google Gemini API + React/Django) in Bronco Hackathon 2025

### Chinese Students and Scholars Association (CSSA), Vice President

January 2024 – May 2026

- Led 300+ member organization; developed Python QR code registration system streamlining event check-ins for 500+ attendees; designed event posters, organization mascot, and digital media content

## SKILLS

- Programming: Python (Advanced), Java, JavaScript/TypeScript, C/C++, Kotlin, SQL, HTML/CSS
- ML/AI: PyTorch, TensorFlow/Keras, scikit-learn, Transformers, LoRA/PEFT, LangChain
- Frameworks: React, Django, FastAPI, Flask, MPI/mpi4py
- Tools: Git/GitHub, Linux, Docker, Pandas, NumPy, Vercel
- Languages: English, Chinese (Mandarin) – Bilingual
- Other: UI/UX design, Digital/Traditional Illustration, Character & Poster Design, Data Visualization